THE CHURCH OF ENGLAND

ETHICAL INVESTMENT ADVISORY GROUP

Artificial Intelligence

The Advice of the Church of England Ethical Investment Advisory Group

December 2024

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The Church of England Ethical Advisory Group (EIAG) provides independent ethical advice on investment to the Church of England's three national investing bodies (NIBs), detailed below.

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Contents

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EXECUTIVE SUMMARY

Few technologies have gripped the public imagination like Artificial Intelligence (AI). The rapid growth and advance of AI technology and applications, notably open access tools such as ChatGPT, have raised awareness of the risks and scale of impact of AI for many aspects of life, including even the fear of dystopian or apocalyptic futures. Leaving aside these more extreme futuristic scenarios, this advice paper focuses on practical applications of AI as they exist today or in advanced development, particularly in the private sector. It builds on the EIAG's previous *Big Tech* report which noted the importance of AI in the surveillance capital business model, but which highlighted that there were ethical concerns with AI that extended beyond the remit of that report.[1]

The term AI is difficult to define, but for the purposes of this report we take it to mean artificial neural network (and related) algorithms that carry out tasks normally performed by humans. As is now widely appreciated, the accuracy or appropriateness of an AI algorithm's output in response to a given input is strongly dependent on the skill and thoroughness by which it has previously been configured and trained by its human designer(s). Although AI can give the (false) impression of real agency and understanding, it can also, if poorly designed or poorly trained, give biased, misleading, or bizarrely unsuitable results. Furthermore, the causes of such defects and malfunctions are not easy to diagnose and correct, owing to the fluidity and opacity of the algorithm's internal architecture. So, while AI—like all technological tools—offers undeniable opportunities for human flourishing, it also carries significant risks. Some of these risks arise from its potential to surpass and replace human performance (e.g. leading to job obsolescence), others from its potential for erroneous or spurious guidance and decision making (e.g. reinforcing societal prejudices), and still others from its potential to be weaponised by bad actors (e.g. for "deepfaking"). AI is now to be found, or at least claimed, in nearly all market sectors, offering benefits that include the improved efficiency and effectiveness of existing offerings as well as radically new innovations. By the same token, some of the risks and ethical implications of AI are similar in kind to those of existing products and services, while others are quite different in kind, requiring fresh thinking and more cautious approaches.

Such concerns have prompted a proliferation of responses from governments, communities of practice and individual businesses in the form of guidelines, codes of practice and new legislation, from which can be gleaned recurring secular ethical values of relevance to AI. These include safety and security, privacy, fairness, human dignity, societal repercussions, transparency and accountability. But rather than adopt them at face value, we have first mapped them against the five key Christian values that were first presented in the *Big Tech* report, namely: flourishing as persons, flourishing in relationship, standing with the marginalised, caring for creation, and serving the common good. In doing so, we have tried to make explicit the biblical and theological basis for some values that might otherwise be taken for granted, such as grounding the right to privacy in the commandment "you shall not steal" or grounding human dignity in the concept of *imago dei*.

To illustrate how these Christian (and secular) values might bear upon AI, we have selected case studies representing two main categories of AI application — "Analytical" or "Generative"—each operating in two different modes—"Non-autonomous" or "Autonomous" in the case of Analytical AI, and "Static/One-Way" or "Dynamic/Interactive" in the case of Generative AI. For each of the four chosen case studies—Medical Diagnostics, Financial Products, AI-generated Art and Scripts, and Robotic Healthcare Companions—we have analysed their operative principles, their social impact, and their relation to Christian values. Inevitably, pros and cons emerge for each, and we comment on how the tensions may be balanced, ameliorated or resolved. We then offer a number of practical considerations for companies offering such products or services (and for investors reviewing their suitability for investment).

The sheer breadth and pervasiveness of AI applications, along with the speed at which AI is developing, create challenges for framing practical investment advice. Where AI is not

1. See EIAG, *Big Tech: The Policy of the Church of England National Investing Bodies and The Advice of the Church of England Ethical Investment Advisory Group*, available at: http://www.churchofengland.org/eiag/policies, ¶11.

adding any new ethical dimensions to an existing application, the reader is encouraged to consult previous EIAG advisory reports. However, with regard to addressing the unique risks associated with AI, the EIAG expects both developers of AI and companies deploying and using AI in their operations to **take moral responsibility** for the additional risks associated with these systems. The EIAG would support efforts to establish industry standards of practice for the whole AI lifecycle and, where necessary, the development of appropriate regulation to support this. Specifically, we would expect to see a commitment to:

i **Human-centred design and deployment**
Companies involved in any part of the lifecycle of AI have a responsibility to ensure that they protect human rights and that AI does not violate these at any stage. AI should not perpetuate bias and discriminate against people based on "protected characteristics", and training data should be geographically and socially representative. Where AI is deployed in a company, managers should ensure employees are given appropriate training to use AI and, where this may make jobs redundant, offer fair and equal opportunities for high-value upskilling and reskilling. Where AI is used appropriately in decision-making, this must include human verification and determination, i.e., "a human always in the loop".

ii **Accountability**
Company leaders developing and deploying AI should take responsibility for ensuring the safety and security of the AI technologies that they either develop or outsource, and be held accountable for any malfunctioning algorithms. The EIAG expects these companies to consider the full impact of AI development, deployment and use. Companies developing AI should clearly define the purpose of the AI system in development and integrate senior management-level accountability for responsible development, training and deployment of AI. Developers should also ensure guardrails are in place to prevent AI systems going beyond their intended remit, and have mitigations to prevent AI being used for unethical or nefarious purposes.

iii **Transparency**
Companies developing AI should be transparent about the sources of data used for developing and training AI, which should not include data subject to copyright (without obtaining appropriate license), or personal or confidential data where explicit consent has not been given. Where AI systems are used in decision making that has a significant impact on an individual's safety, rights or wellbeing, AI should be explainable and auditable. Companies deploying AI should be transparent about when and how they are using AI, including signposting means of appeal and redress when AI is used in decision making. Products of generative AI systems should contain means to verify whether AI has been employed in their creation.

INTRODUCTION

1 Artificial intelligence (AI), once a science fiction fantasy, is now inexorably part of the world's social fabric. Rapid growth and advancement of AI technology and applications, notably open access tools such as ChatGPT, have propelled conversations about AI safety and ethics into the public sphere. Increasing awareness of the risks and the potential scale of impact, together with a lack of established global norms on the development and use of AI, has prompted a proliferation of ethical responses from institutions and communities of practice. That said, it should be borne in mind that there is currently also much "hype" about the potential of AI, which is reflected in the dominance of AI and tech-based companies in current markets, and it is unclear the extent to which this technology will fulfil the extremes of these expectations.[2]

2 The Church of England's General Synod agreed in 2024 to endorse the Rome Call for AI Ethics, a call signed by the Pontifical Academy for Life in Rome, together with leading technology

2. The Economist, "Artificial Intelligence is losing hype" (19 August 2024), available at: https://www.economist.com/finance-and-economics/2024/08/19/artificial-intelligence-is-losing-hype.

firms and the Italian Government's Ministry of Innovation.[3] The Call comprises three impact areas: ethics, education, rights, and six principles: transparency, inclusion, accountability, impartiality, reliability, and security and privacy.

3 The EIAG's *Big Tech* report noted the importance of the use of AI in the attention economy business model, and highlighted that there were wider ethical concerns which extended beyond the remit of that report.[4] This report aims to address these wider concerns, to offer a distinctively Christian (and Anglican) perspective on the ethics of AI, and to support the National Investing Bodies to act as Christian investors in relation to their investments and stewardship in matters concerning the development and deployment of AI. The advice is informed by a close attention to the biblical and theological background to the EIAG's theological principles that first appear in the *Big Tech* report.[5]

The scope of this advice

4 This advice paper will focus largely on practical applications of AI as they exist now or in advanced development—particularly in the private sector—by either developers or users, rather than on futuristic scenarios such as general-purpose AI and the possibility of a malevolent conscious AI.

5 The advice of the EIAG is biblically and theologically grounded. Our thinking is formed around a framework of five principles which affirm that human beings live well and are fulfilled when they are:

i. flourishing as persons;
ii. flourishing in relationship;
iii. standing with the marginalised;
iv. caring for creation;
v. serving the common good.

6 This framework brings to bear various ethical principles, some of which are generally applicable to all technology and some of which are more specific to AI and need greater elucidation.

7 While the EIAG seeks to develop advice that has broad application to ensure advice remains relevant, the pace at which AI technologies are developing and pervading public life and business activities presents a challenge for creating enduring ethical investment advice. We recognise this advice (published in December 2024) will not be the EIAG's final word on AI, with future developments in AI not yet taken account of. It will no doubt be necessary to revisit this advisory paper with greater frequency to ensure the advice and its recommendations remain relevant and practical for investors.

Structure of this advice

8 This paper will begin by exploring the definition of AI and potential future developments of AI technology and some of the challenges inherent in addressing ethical questions about AI. The second section of this advice is a theological and ethical reflection on AI technology, within the scope of this paper, grounded in the EIAG's theological framework. To elucidate these ethical reflections, we will then consider four case studies of different AI categories and applications and the specific ethical considerations these raise. Finally, on the basis of these theological and ethical reflections, we will propose ways in which investors can engage with companies and public policy to support responsible and ethical use of AI.

WHAT IS AI?

9 The term "artificial intelligence" (AI) was coined in 1955 by early computer scientist John McCarthy, to describe a research project to simulate intelligence in a computer by codifying

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3. The Church of England, "Synod affirms work as key to 'human dignity and purpose' in the face of the AI revolution" (26 February 2024), available at: https://www.churchofengland.org/media/press-releases/synod-affirms-work-key-human-dignity-and-purpose-face-ai-revolution. More information about the Rome Call for AI Ethics is available at: https://www.romecall.org/.
4. EIAG, *Big Tech,* ¶11.
5. EIAG, *Big Tech,* pp. 17–20.

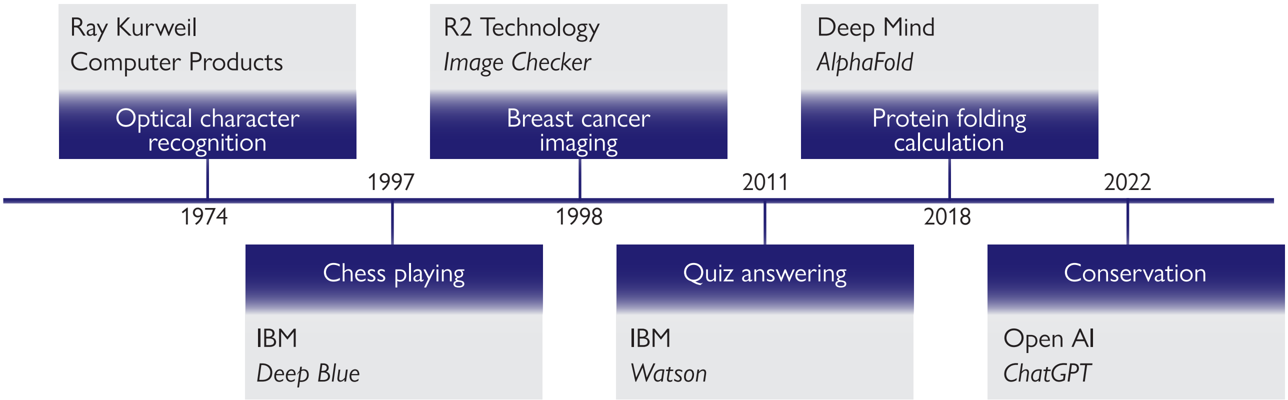

Figure 1 — Notable milestones in AI

language and general problem-solving logic.[6] While resisting precise definition,[7] the term has been used to describe any application of computers to tasks which might normally be performed by a human person. This broad coverage makes separating AI from regular computing all but impossible, partly because of the breadth of techniques employed by AI,[8] and partly because, as computers demonstrate success in human endeavours such as chess and natural language processing—and even beyond-human tasks such as predicting how proteins fold—these achievements (overviewed in Figure 1) can be seen as the result of comprehensible computer programs and not evidence of artificial intelligence *per se*.[9]

10 At the heart of the current generation of AI applications lies one key technology: simulated neural networks. These systems work by simulating the way in which the biological brain operates, rather than by explicitly coding for each foreseeable outcome. The relative weightings of connections can be adjusted in response to trial-and-error scenarios to optimise the network to produce the desired result—and to continually adapt the network throughout its usage.[10] The combination of the vast power of today's computers, together with the huge amount of digitised data which is available to train these networks, has enabled the production of attention-grabbing systems like OpenAI's ChatGPT. One unique feature which AI enables is mimicry of human actions and reactions, which gives the illusion of agency and intent on the part of the AI system itself.

11 Given that these networks are, to a large extent, trained rather than designed, several common issues arise: the lack of transparency in the

6. For a succinct summary of the early history of AI, see Noreen Herzfeld, *The artifice of intelligence: divine and human relationship in a robotic age* (Minneapolis, MN: Fortress Press, 2023), pp. 3–8. Other accessible summaries include: Margaret A. Boden, *Artificial intelligence: a very short introduction* (Oxford: Oxford University Press, 2018); Eve Poole, *Robot souls: programming in humanity* (Boca Raton and Abingdon: CRC Press, 2024); Tom Taulli, *Artificial intelligence basics: a non-technical introduction* (Monrovia, CA: Apress, 2019).

7. Examples of definitions of AI include: "The science of making machines do the things that would require intelligence if done by men"—Marvin Minsky, "Preface", in *Semantic information processing*, ed. Marvin Minsky (Cambridge, MA and London, UK: The MIT Press, 1968), p. v; "The pursuit of computer programs that are capable of reproducing some aspect of human cognition"—David Gelernter, *The muse in the machine: computerising the poetry of human thought* (New York: Free Press, 1994), p. 44; "Technologies with the ability to perform tasks that would otherwise require human intelligence, such as visual perception, speech recognition, and language translation"—Select Committee on Artificial Intelligence, *AI in the UK: ready, willing and able?* (2018), p. 14; "AI is simply the display by a machine of any cognitive process that we would expect to be done by a person"—Peter Robinson, "Introduction: a computer technology perspective", in *The robot will see you now: artificial intelligence and the Christian faith*, ed. John Wyatt and Stephen N. Williams (London: SPCK, 2021), p. 3.

8. These include: machine learning, deep learning, reinforcement learning, natural language processing, collaborative systems, computer vision, algorithmic game theory, robotics.

9. For an example of a high-bar approach to AI, see: Melanie Mitchell, "AI's challenge of understanding the world", *Science* 382, no. 6671 (10 November 2023).

10. Using computers to simulate neural networks is not new—Marvin Minsky and Dean Edmonds demonstrated this in 1950 with their 40-neuron SNARC machine. See Martijn Kuipers and Ramjee Prasad, "Journey of artificial intelligence", *Wireless Personal Communications* 123, no. 4 (April 1 2022).

system's decision making; the propensity for poor or biased training data to result in similarly poor or biased performance in the real world; and the potential for seemingly innocuous inputs resulting in truly bizarre responses as a result of an untested path through the network being followed.

12 AI *applications* should be distinguished from AI *algorithms*. Most AI applications will be found in products and services that have been in existence well before the arrival of AI technology. The developers of these products and services are simply incorporating AI to enhance their offerings in some way, making them more accurate, efficient or reliable, or perhaps adding new features and benefits that have previously been unavailable. Examples include the use of AI to make more accurate decisions about to whom to offer financial products, or to make more accurate diagnoses of medical images captured by CT or MRI. Some AI applications may be entirely new, having been brought about by the arrival of AI. The proliferation of smartphones and social media has, for example, enabled the collection of vast quantities of personal data that can be used to train AI algorithms for the purpose of profiling users in order to offer them bespoke products and services, and this is explored in the EIAG's *Big Tech* report.[11] Here the collection of data and AI go hand in hand, for without AI the data would be difficult to analyse, and without the training data the AI would be inaccurate. Some institutions and businesses, however, are developing AI algorithms *per se*, either with a view to marketing these as licensable software, or with a view to enhancing our understanding of human or artificial intelligence. Examples of AI markets and applications can be found in Appendix 1.

13 The focus of this advice lies at the intersection of the market applications and the algorithms. Particular attention should be given to traditional products and services that we take for granted but which have incorporated AI. This will not always be announced. Some developers might smuggle AI into their product or service with the hope that it will go unnoticed by the customer. Others might use the new AI as a marketing opportunity to boost sales. For this reason, it is important for companies to understand where AI is used within their existing operations and to seek clarity on the modes of operation and use of AI in new technologies during the procurement phase.

14 For the purposes of this report, and to keep it within reasonable bounds, we propose a scope that focuses on what is today commonly regarded as AI, without pulling in old well-established technology and without entertaining speculative futuristic technology which has little bearing on today's investment opportunities. To avoid repetition, we will also be bracketing out the AI that has already been scrutinised as part of the EIAG *Big Tech* report. Furthermore, it is important to distinguish between AI *per se* and many adjunct technologies that AI often relies upon, such as computers, the internet, telecommunications, social media, etc. It should be borne in mind that some of the ethical and societal issues arising from AI may have been created first by an adjunct technology and only later exacerbated by AI. A case in point are the political harms caused by social media that have been greatly amplified by Big Data and AI.[12]

15 We propose bracketing out old technology and futuristic technology by means of two criteria: (1) the AI technology must be utilising some form of machine learning neural network rather than just performing static analytical calculations, and (2) the AI in question must either exist today as a usable technology, product or service, or must be in an advanced stage of development.

16 However, the EIAG believes that ethical considerations about AI should relate to the full product lifecycle. Therefore, it is worth being

11. EIAG, *Big Tech,* ¶61–92.

12. Philipp Lorenz-Spreen et al., "A systematic review of worldwide causal and correlational evidence on digital media and democracy", *Nature Human Behaviour* 7 (2023).

aware of some future technologies that might be on the radar screen of certain technology developers and that raise some very serious theological and societal concerns. For example:

17 *Conscious Artificial Intelligence (CAI) and Artificial General Intelligence (AGI).* If AI could become sentient in a way that approximates our own thought processes or self-consciousness as human beings, this would raise many questions. Should such CAI or AGI be treated as a moral agent and/or a moral patient, that is to say, morally responsible for its actions and deserving of moral rights? Should we treat it as an equal partner on earth or as a potential threat that needs to be contained at the outset? And how would we ever know that the consciousness is genuine, given the way much AI has been designed to simulate and deceive?[13] In any case, CAI might not emerge abruptly but might go through multiple transitional states; in which case, what should our *modus operandi* be towards machines that merely emulate human consciousness? Even now, some of these intermediate states may not be too far away, and we have a choice as to whether to be civil or rude to a virtual ChatBot or an embodied robot.[14] There is no shortage of theological and philosophical literature on the subject of CAI, but most of it goes beyond the scope of this report.[15]

18 *Transhumanism and posthumanism*. These technological and philosophical movements seek eternal life, or a life enhanced beyond what would be considered to be naturally human.[16] Routes to achieving this include "digital immortality" (e.g. uploading our minds onto a computer),[17] and cyborg enhancement (e.g. grafting machines into or onto our bodies).[18] Strictly speaking, neither of these routes need involve AI algorithms of the kind discussed in this report; however, to the extent that they prolong or enhance human intelligence (however that is defined), they may be said to be creating a new form of human intelligence that is to some extent "artificial". It is worth being aware of these futuristic technologies since they capture the public's imagination and have led to much popular and academic discussion.[19]

CHALLENGES IN ADDRESSING AI

Pervasiveness

19 Whilst public attention on AI has dramatically increased, AI itself, in the form of embedded

13. Note the recent case of the Google engineer, Blake Lemoine, who claimed that Google's LaMDA was conscious and sentient.
14. For theological reasons to treat non-sentient technology as a "you" rather than an "it", see Michael S. Burdett, "Personhood and creation in an age of robots and AI: can we say 'you' to artefacts?", *Zygon: Journal of Religion and Science* 55, no. 2 (2020).
15. As an introductory sample, see: Kate Darling, "'Who's Johnny?' Anthropomorphic framing in human-robot interaction, integration, and policy", in *Robot ethics 2.0: from autonomous cars to artificial intelligence*, ed. Patrick Lin, Ryan Jenkins and Keith Abney (Oxford: Oxford University Press, 2017); Simon Balle, "Theological dimensions of humanlike robots: a roadmap for theological enquiry", *Theology and Science* 21, no. 1 (2023); Randy Beavers et al., "Technology and non-interpersonal relationships", *Journal of Biblical Integration in Business* 23, no. 1 (2020).
16. For an introduction, see: Robert Ranisch and Stefan Lorenz Sorgner, "Introducing post- and transhumanism", in *Post- and transhumanism: an introduction*, ed. Stefan Lorenz Sorgner and Robert Ranisch (Frankfurt, Germany: Peter Lang, 2014).
17. While the transferring of our minds (and "lives"?) to a different hardware platform is still the stuff of science fiction, the possibility of our "living on" for the benefit of bereaved family and friends through "digital necromancy" AI programmes which resurrect us into virtual persons is much closer. See: Maggi Savin-Baden and David Burden, "Digital immortality and virtual humans", *Postdigital science and education* 1 (2019); James Hutson and Jeremiah Ratican, "Life, death, and AI: exploring digital necromancy in popular culture—ethical considerations, technological limitations, and the pet cemetery conundrum", *Metaverse* 4, no. 1 (2023), available at: https://doi.org/10.54517/m.v4i1.2166.
18. A current example of this is Elon Musk's Neurolink brain-computer interface research.
19. Some good discussions for and against can be found in: Ronald Cole-Turner, "Transhumanism and Christianity", in *Transhumanity and transcendence. Christian hope in an age of technogical enhencement.*, ed. Ronald Cole-Turner (Washington DC: Georgetown University Press, 2011); Michael Burdett and Victoria Lorrimar, "Creatures bound for glory: biotechnological enhancement and visions of human flourishing", *Studies in Christian Ethics* 32, no. 2 (2019); Nick Bostrom, *Superintelligence: paths, dangers, strategies* (Oxford: Oxford University Press, 2014); Douglas Estes, "Postdigital humans: technology and divine design", in *Postdigital theologies: technology, belief, and practice*, ed. Maggi Savin-Baden and John Reader (Cham, Switzerland: Springer, 2022); Jeanine Thweatt-Bates, *Cyborg Selves: A Theological Anthropology of the Posthuman* (Farnham: Ashgate, 2012).

firmware or application software, has been in existence for decades. AI algorithms *per se* are in the public domain and there is nothing to stop anyone from writing their own code (except lack of expertise, computer hardware, or the raw data to do the training). More recent developments are toolkits that enable users to customise AI to their own domain.[20]

20 Although many companies might want to take advantage of the (generally) popular image of AI to market their product, it is quite easy to conceal from the customer the fact some kind of AI is employed in the product and AI may be more widely deployed than we realise.[21] This represents a particular challenge from the outset for the NIBs as investors—the availability of data about the nature and extent of companies' use of AI.

Market exaggeration

21 There have already been at least two cycles of popularity and decline in the history of AI and there is no guarantee that the current fascination with AI will last. AI is currently capturing public imagination such that any claim to be using AI is likely to boost the publicity behind a new product. The wide range of meanings that AI can have makes it easy for developers to claim that AI is used in their product, even if there are no AI techniques actually involved, without fear of breaching any trade description laws.

Parsing generic versus specific aspects of AI

22 Like any technology, AI can be designed/intended for good or bad purposes. Those same purposes, good or bad, might often be achievable by other technologies. According to this view, it is not AI *per se* that should be under scrutiny but the product, service or technology application of which it is a part. It could be claimed that it is unfair to single out those products incorporating AI for ethical evaluation and better to review according to market application or business model rather than technology. Here the focus is on the agent and the agent's goal rather than the instrument (e.g. AI) that the agent uses. In other words, ethical questions on the deployment of AI are context dependent. For example, as highlighted in the *Big Tech* report, the use of AI in the collection and use of personal data is not inherently a bad thing until issues concerning privacy and informed consent are laid bare.[22]

23 However, there are times when AI makes possible new purposes or better (e.g. more efficient) ways of accomplishing existing purposes, in which case the AI technology itself carries ethical freight and deserves separate consideration (though the US National Rifle Association's slogan "Guns don't kill people, people do" is an example of how this point can be contested).[23]

24 Also, if AI is uniquely capable of producing serious unintended consequences, then again, it is worthy of separate ethical consideration, even if its intended purposes are no different from those of many other technologies. Foremost among such possible unintended consequences are those resulting from its alleged autonomy and potential unpredictability. Despite thorough training and algorithm design, there have been cases of AI making serious mistakes when put to use in the real world.

25 On the subject of "mistakes", insofar as AI incorporates empirical/contingent data and uses statistical/probabilistic algorithms it is no different from all other technologies that utilise the same: all such inductive technologies are likely to achieve less than 100% accuracy in their predictions (unlike purely deductive technologies such as pocket calculators that are merely carrying out mathematical transformations). Like any inductive technology, we should expect AI to occasionally make mistakes even if it is generally much more accurate and reliable than a human. It would be to have ethical double

20. Open AI, "Introducing GPTs" (6 November 2023), available at: https://openai.com/blog/introducing-gpts.
21. See Mark Coeckelbergh, *AI ethics* (Cambridge, MA and London: The MIT Press, 2020), p. 78.
22. See EIAG, *Big Tech*, ¶101–104.
23. Herzfeld, *The artifice of intelligence*, p. 98, fn.31.

standards if, for example, AI-based weather forecasting were to receive greater ethical scrutiny than non-AI-based forecasting, simply because it contained AI.

26 Even so, if the *kind* of mistakes that AI can make in certain products and services are qualitatively very different from those made by allied technologies then it would be appropriate to earmark such products and services for greater ethical scrutiny.

27 An additional concern, arguably unique to AI, is its ability to deceive us as to its own nature, to appear to be a person rather than an object or instrument. Such impersonation may be the intent of the designer (e.g. when an AI chatbot is substituted for a human being in a live chat helpline without letting the user know), or inadvertent (e.g. when a care home patient starts to believe that a clearly labelled robot assistant is really a human being). Either way, the psychological or social consequences of such mistaken identity may become quite significant (as detailed below).

THEOLOGICAL AND ETHICAL REFLECTION

28 AI ethics is lagging behind its implementation. As Eve Poole notes, "Apart from generic ethics committees, AI was left untouched for years, until there were scandals about the military use of un-manned drones, the bias in algorithms, and the abuse of private data. So in the emergent regulatory statements and frameworks it is possible to identify several doors being locked after horses have long since bolted."[24] There is now a substantial literature on the ethics of AI, in which some recurring themes or principles can be found.[25]

29 The framework of five principles around which the EIAG's thinking is formed helps to identify preliminary ethical considerations raised by AI in general:

Principle	Ethical concern
Flourishing as persons	• That it does not undermine inherent human dignity • That it does not lead to a devaluation of good work and of human skill, expertise and creativity
Flourishing in relationship	• That it does not lead to a devaluation of existing human relationships (i.e. by getting too used to the attenuated relationships afforded by robots) • That it encourages honesty and authenticity (e.g. AI is clearly labelled as such to minimise wilful or inadvertent deception)
Standing with the marginalised	• That it does not exacerbate societal prejudice (e.g. by training AI algorithms with biased data sets)
Caring for the creation	• That it is not wasteful of resources (e.g. needlessly encouraging single use or constant upgrade to new product models) • That it does not add significantly or needlessly to emissions (e.g. through the need for more energy-consuming data servers)
Serving the common good	• That it does not lead to greater inequality through poor distribution of the benefits of AI

24. Poole, *Robot souls*, p. 17.
25. The following is a small sample: Mark Coeckelbergh, *AI ethics* (Cambridge, MA: MIT Press, 2020); Institute of Business Ethics, "Business ethics and artificial intelligence", *Business Ethics Briefing*, Issue 58 (January 2018); Luciano Floridi and Josh Cowls, "A unified framework of five principles for AI in society", in *Ethics, governance, and policies in artificial intelligence*, ed. Luciano Floridi (Cham, Switzerland: Springer, 2021); Nuffield Foundation, *Ethical and societal implications of algorithms, data, and artificial intelligence: a roadmap for research* (London, 2019); Matthew J. Gaudet, "An introduction to the ethics of artificial intelligence", *Journal of Moral Theology* 11, Special Issue 1 (2022); S. Matthew Liao, "A short introduction to the ethics of artificial intelligence", in *Ethics of artificial intelligence*, ed. S. Matthew Liao (Oxford: Oxford University Press, 2020); Jessica Morley et al., "The ethics of AI in health care: a mapping review", in *Ethics, governance, and policies in artificial intelligence*, ed. Luciano Floridi (Cham, Switzerland: Springer, 2021); Poole, *Robot souls*; UNESCO, *Recommendations on the ethics of artificial intelligence* (Paris: UNESCO, 23 November 2021).

30 A further consideration—which speaks to speculative and existential concerns about AI—is the ontological limits to human stewardship and whether the mandate given to humanity entitles us to create new sentient life forms that might eventually experience real suffering and/or be morally culpable.

Secular ethical values

31 One of the most common ways of articulating ethical values in a secular context is through the language of human rights. The EIAG's advice on human rights and the resulting NIB policies aim to prevent or mitigate the risk of adverse impacts on human rights through good stewardship of business relationships and investments.[26] Human rights are essentially the fundamental rights and freedoms of all human beings and are based on the values of dignity, fairness, respect and justice for all, regardless of nationality, gender, race or ethnicity, religion, language or other relevant characteristics.[27]

32 These human rights need to be interpreted and applied more specifically in different contexts, and AI is no exception.[28] Accordingly, several ethical values and considerations tend to recur in governmental and institutional reports relating to AI (e.g. as summarised in the Nuffield Foundation report[29]). These are similar to those in the Rome Call which have already noted in this paper but we have chosen the following values for analysis given that these originate from a cross-section of reports. These include the following:

- safety, security
- privacy
- fairness, non-discrimination
- human dignity
- societal repercussions (e.g. job deskilling and obsoleting, personalisation leading to loss of societal solidarity and risk pooling)
- transparency, explainability
- accountability, responsibility

33 These secular values may be mapped onto the Christian values above, as shown in Figure 2. Whilst mapping the values in this way is simplistic and risks losing their nuance, this is intended to facilitate an understanding of how these two values frameworks overlap.

34 **Safety and security** are important prerequisites for human flourishing, whether that be personal, relational, societal or global. This is true in two senses. First, when, due to inadequate safety and security, harmful events take place, our flourishing as individuals, communities and ecosystems will obviously be adversely affected. Second, even the awareness of unsafe and insecure environments may create anxiety, suspicion and reticence that can fragment society, leading to lack of trust and an inward-looking mentality. For example, we might fear that powerful AI lacking in adequate security might be taken over by terrorists, or might itself take over the world, but even if those scenarios never materialise, inadequate protections on the part of developers of AI can lead to much societal unease, which in turn can lead to reputational damage and a loss of a social license to operate for companies developing and using AI.

35 The concept of **privacy** has complex historical and cultural roots and is difficult to define. Attempted definitions include the right to be left alone, the right to appropriate flow of personal information, and the right to conceal and safeguard personal informational property.[30] Concepts of human personhood that emphasise the importance of relationality in framing or

26. EIAG, *Human rights: the policy of the Church of England National Investing Bodies and the advice of the Church of England Ehtical Investment Advisory Group*, available at: http://www.churchofengland.org/eiag/policies.
27. EIAG, *Human rights*, p. 7.
28. The United Nations Human Rights Commission has published some useful material on AI. See OHCHR, "B-Tech Project: OHCHR and business and human rights", available at https://www.ohchr.org/en/business-and-human-rights/b-tech-project, and in particular UHCHR, "Taxonomy of Human Rights Risks Connected to Generative AI", available at https://www.ohchr.org/sites/default/files/documents/issues/business/b-tech/taxonomy-GenAI-Human-Rights-Harms.pdf.
29. Nuffield Foundation, *Ethical and societal implications.*

30. Beate Roessler and Judith DeCew, "Privacy", in *The Stanford Encyclopedia of Philosophy*, ed. Edward N. Zalta and Uri Nodelman (2023), available at: https://plato.stanford.edu/archives/win2023/entries/privacy/.

		CHRISTIAN VALUES				
		Flourishing as persons	Flourishing in relationship	Standing with the marginalised	Caring for creation	Serving the common good
SECULAR VALUES	Safety / security	***	**	**	**	**
	Privacy	**	***	*	*	*
	Fairness / non-discrimination	**	**	***	*	**
	Human dignity	***	***	***	*	**
	Societal repercussions	*	**	**	**	***
	Transparency / explainability	*	***	*	*	**
	Accountability / responsibility	**	***	**	**	**

KEY: * weak bearing ** medium bearing *** strong bearing

Figure 2 – Secular values mapped onto Christian values

constituting the person might sit uneasily with overly strong claims to human privacy, i.e. too much privacy may be unhealthy. However, too little privacy can also erode our freedom and autonomy. Perhaps the best approach to take on privacy from a Christian perspective is not to emphasise human rights to individual autonomy and isolation *per se*, but to emphasise the duty of all to respect the wishes of others, on the presumption that a person's wishes are in line with their flourishing as persons.[31] Of course, this is not always the case, in which case forceful invasion of privacy might be warranted.

36 Another possible angle is to treat privacy as ownership of personal informational property, the infringement of which is effectively to break the 8th/9th commandment "thou shalt not steal". AI developers and users should therefore be cognisant of the tendency for their technology to impact on relational and individual flourishing by deliberate or unintentional invasion of privacy. To avoid such invasion, consent is needed, but paying lip service to consent (e.g. simply asking for a box to be ticked after lots of small print) is not the same as gaining *fully-informed* consent (i.e. after ensuring people understand how their data are being used).[32] In whichever way we may prefer to ground privacy in biblical principles, and no matter how much the right to privacy needs to be balanced against other civic duties, it is likely to be the case that some boundaries against egregious infringement of privacy need to be set.

37 **Fairness or non-discrimination** is a clear biblical principle enshrined in injunctions such as not to neglect the orphan and widow, not

31. Nathan Mladin and Stephen N. Williams, "The question of surveillance capitalism", in *The robot will see you now: artificial intelligence and the Christian faith*, ed. John Wyatt and Stephen N. Williams (London: SPCK, 2021), p. 225.

32. For more discussion on this topic, see: EIAG, *Big Tech*, ¶61–67.

to take bribes, and not to use dishonest scales. Fundamentally, the call for social justice lies in our spiritual equality before God as human beings made in his image, irrespective of any material inequalities that might subsequently befall us. Although in theory this principle is commonly regarded as self-evident in western democracies, in practice that is not always the case, especially for those less able to claim and assert their rights to equal treatment and equal benefits. For this reason, in a Christian context, the principle is underpinned by a preferential option for the poor. Discrimination might also be inherent in the fact that much of the data used to train AI is from the USA, Europe and developed economies, thus ignoring a vast majority of people, their experiences and preferences.

38 Job obsolescence due to AI is a real concern, particularly if it happens without the opportunity for a just transition. Companies and governments have a responsibility to plan and prepare for workforce displacement and to support workers through meaningful reskilling opportunities for those affected and through transitioning education and training of younger generations to reflect the skills and expertise needed for a future workforce.

39 One issue that AI could exacerbate is the concentration of wealth in a very small elite. The development and participation in AI technologies should be inclusive, ensuring that the benefits of AI technologies are accessible and valuable to all. The UNESCO recommends that "[t]he most technologically advanced countries have a responsibility of solidarity with the least advanced to ensure that the benefits of AI technologies are shared, such that access to and participation in the AI system lifecycle for the latter contributes to a fairer world order with regard to information, communication, culture, education, research and socio-economic and political stability".[33] Essential to this is also ensuring that data ecosystems do not reinforce existing inequalities and power dynamics. New frameworks to support integration of data equity are fundamental to ensure that AI systems are inclusive and representative of different geographical, cultural and social communities so that marginalised groups can also meaningfully benefit from AI systems.[34]

40 **Human dignity** is the belief that people hold a special value tied to their human nature, and is the ultimate basis for many of the other secular values. It accounts for why we do not treat human persons in the same way we treat nonhuman organisms, machines, or rocks. For most of the time we know what is human and what is not, and what bestowing dignity means, that is to say, showing respect, cherishing, protecting, seeking to understand and to love. In our transactional economic world, it is easy to lose sight of the fact that the economic agents in question are human persons whose dignity needs to be maintained irrespective of economic gain. Human dignity relates again to the theological concept of the *imago dei*. While we do not deserve to be worshipped, we do deserve to be given the full respect that is due to our being in relationship with God, our representing him as viceregents on earth, and our participating in some of his attributes.

41 The advent of AGI and CAI would raise the prospect of whether human dignity should ever be conferred to machines that think and act indistinguishably from human beings. But closer to the here and now, the way we interact with AI-powered machines and avatars might also unintentionally affect our human dignity, since relationships with machines are likely to be poor substitutes for relationships with real human beings. Over long periods of time too much accommodation or "dumbing down" to machines might cause atrophy of our personhood and self-esteem. Although these fears might seem exaggerated, it is

33. UNESCO, "Recommendation on the Ethics of Artificial Intelligence" (2022), available at: https://www.unesco.org/en/articles/recommendation-ethics-artificial-intelligence.

34. World Economic Forum, "Data Equity: Foundational Concepts for Generative AI" (October 2023), available at: https://www.weforum.org/publications/data-equity-foundational-concepts-for-generative-ai/.

worth considering that AI, unlike most other technology, has the unique potential to engage us in relationship, and since flourishing in relationship is key to individual flourishing, we should ever bear in mind what AI might be doing *to* us, as well as what AI is clearly doing *for* us.

42 Attending to flourishing as persons and in relationship does not, in secular thinking, automatically entail serving the common good and this is why **societal repercussions** are an important consideration in the implementation of technology such as AI. We cannot rely on societal harmony emerging spontaneously from personal and interpersonal flourishing, especially if such flourishing and the virtues that give rise to them are not widespread. A top-down perspective is needed as well as a bottom-up perspective. The Old Testament certainly has much to say about the need for civic laws and regulations, while Jesus and the New Testament writers are not lacking in positive things to say about the need for maintaining societal stability through respect for authorities, taxation, and governance.

43 Myths of capitalism's "invisible hand"[35] can encourage the mistaken belief that it is not the technology provider's responsibility to worry about any harmful societal repercussions that might come about through adoption of their technology. It may indeed be difficult for a single entrepreneur to predict the more subtle and far-reaching unintended consequences of their innovations, but this is where cooperation, explainability and transparency can anticipate and mitigate potential harmful outcomes. Professional institutes, industry advisory bodies and independent think-tanks can all play their part in bringing to bear a collective wisdom that can span politics, sociology, economics, psychology, religion, and much more.

44 **Transparency and explainability** are ethical concerns that arise from the way in which most AI algorithms work. The training of neural networks that leads to the setting of nodal weightings is a hugely complex process that is difficult to reverse engineer (similar to the difficulty of retrospectively analysing all the forces that led to a stone landing at one particular position after falling down a mountainside). When things go wrong with AI—when they fail to make correct judgements or when their judgements are wildly inaccurate—the inscrutability of the algorithm can be an impediment to learning how to improve and avoid future mistakes. At first sight it is difficult to see how this has any precedent in Christian ethics. Perhaps, like the need for safety and security, it would be a dereliction of duty not to take reasonable care to ensure that the tools we use are fit for purpose, especially when they are known to have caused harm on previous occasions. Here one could cite the culpability of a person who fails to restrain the ox that kills, when in the past it has gored someone (Exodus 21:28–29). Accidents do happen, but failure to take precautions against known risks is failure in our duty to enable flourishing in relationship. Insofar as AI developers need transparency and explainability to help mitigate such risks, they should be designing these features into their AI algorithms by default.

45 Closely related to the previous secular value is **accountability and responsibility**. When AI goes wrong, who or what is responsible and who or what is ultimately accountable? The goring ox of the Old Testament was destroyed, but what should be the equivalent fate of an AI that has caused serious harm? Perhaps not deletion of the software, but re-training and bug-fixing. Clearly, a lot depends on whether the AI is operating as the manufacturer and user intended. The manufacturer of a rifle is not responsible for its being used by a terrorist to murder innocent people, but the manufacturer of a rifle that regularly backfires and injures its user would be justifiably liable for selling a defective product. In some cases, AI may be being used for nefarious purposes, and it is the user rather than the developer of the AI who

35. Popularised by the eighteenth-century philosopher and economist Adam Smith, the invisible hand is the idea that if you work for your own selfish interests, the incentives of free markets lead you to act in the interest of wider society, even if this is not something you intended. The term "invisible hand" can apply more generally to any individual action that has unplanned, unintended consequences.

must be held accountable. In other cases, the AI algorithm itself might be poorly designed, irrespective of the uses to which it might be put. From a Christian perspective, one could point to the difference between "high-handed" sin (e.g. egregious defiance against God for selfish gain) versus inadvertent, careless sin (e.g. reckless action without due circumspection). Both are reprehensible, though the latter might escape notice more often and yet be more prevalent. In either case, it is important that accountability for AI systems and their use originates with an individual or legal entity.

46 No-one would deny that AI, like any technology, also brings many benefits or goods (e.g. accuracy, quality, efficiency, personalisation). These goods contribute positively to the Christian and secular ethical values mentioned above.

47 Not commonly reflected in secular ethical frameworks for AI are considerations about the environmental cost of developing, training and using AI, and in particular the high energy consumption of data centres used for developing and training AI systems with very large amounts of data.[36] Other considerations include the emissions and raw materials associated with production of Graphics Processing Units (GPUs) needed to support AI systems. See the EIAG's Climate Change advice (and forthcoming Environment advice) for guidance on this issue.[37]

48 When technology impacts ethical principles both positively and negatively, tensions are created. Addressing and resolving these tensions is, in the language of Christian theology, a matter of *prudence* and *wisdom*. It can involve:[38]

- distinguishing true dilemmas from false ones (e.g. when values themselves are in opposition *versus* when only practical considerations stand in the way of resolving the tension);
- prioritising values to weaken a true dilemma;
- balancing costs and benefits (but not just on the basis of financial gain/loss);
- acting on data (rather than on hearsay);
- practising adaptive management;
- applying the precautionary principle (i.e. playing safe by postponing or foregoing the introduction of a new technology).

CATEGORIES OF AI APPLICATION

49 To enable greater reflection on specific ethical challenges of AI, and to aid the NIBs in identifying or prioritising industries or applications which may require attention, it is helpful to categorise the different operational uses of AI.

50 The following is just one way of classifying AI; it is not a perfect scheme, but is designed to capture the full breadth of AI embodiments and to bring to light as diverse a range of ethical issues as possible. We suggest two main categories of AI, each subdivided into two modes of implementation:

1. **Analytical AI**

   A. Non-autonomous decision-making mode, for example:

      - offering bespoke products and services based on an individual's pattern of internet usage (left to the individual to decide whether to purchase or not);
      - on demand provision of news, commentary and analysis (left to the user how to respond to the information);
      - medical diagnostics (left to the physician to make a decision whether to accept result and take action or not).

36. See Generation Investment Management LLP, "Insights 20: Is AI Sustainable?" (26 November 2024), available at: https://www.generationim.com/our-thinking/insights/is-ai-sustainable/.
37. See EIAG, *Climate Change: The Advisory Paper of the Ethical Investment Advory Group of the Church of England,* available at: http://www.churchofengland.org/eiag/policies.
38. This section draws heavily upon the excellent discussion to be found in: Nuffield Foundation, *Ethical and societal implications*.

B. In autonomous decision-making mode, for example:

- autonomous cars (visualise environment and decide when and where to steer, apply brakes, etc.);
- autonomous weapons (visualise environment and decide whether to fire or detonate);
- financial products in which the approval decision is made automatically by AI on basis of personal data profiling.

2. Generative AI

A. In static, one-way mode, for example:

- AI-generated art (e.g. prose, poetry, music, painting, photography);
- AI-generated scripts (e.g. job applications, exam answers, technical reports).

B. In dynamic, interactive mode, for example:

- robotic companions;
- virtual world games;
- virtual reality relationships (e.g. avatars of deceased relatives).

Benefits of this approach

51 Classifying AI in this way helps to expose possible differences in value and risk associated with how AI is used. It might prompt us to ask:

- Is using Analytical AI more legitimate and valuable than Generative AI? In the former, AI might be said to be augmenting human capabilities (solving problems we cannot solve), but in the latter AI might be said to be replacing humans altogether (but without enhancing [the best of] their capabilities), though proponents of generative mode might argue that it improves life by stimulating people to greater productivity.
- Does using AI without autonomous decision making (1A) or without the ability for continuous dialogue with the user (2A) raise fewer issues than with decision-making capability (1B) or conversational ability (2B)? In the former case there is little risk of confusing the instrument as an agent since it is not talking back. One of the consequences of treating an "it" as a "thou" is that, in time, we might ourselves atrophy to a more "it-like" state through habitual acceptance of lower standards of relationship.

Remaining focus of this report

52 For the rest of this advice, we will consider in more detail one representative case study for each of the four categories of AI application.

53 The case studies have been selected primarily to illustrate the diversity of ethical issues that can be encountered with AI, rather than those currently most frequently encountered, with the intention that this advice might be as enduring as possible for the NIBs in assessing future uses of AI. To that end, the EIAG emphasises the importance of the ethical principles and Christian values at stake more than the particular product or technology embodiments. For example, the case study chosen in Section 10 for Generative AI in static/one-way mode (AI-generated art and scripts) may not represent a significant current investment opportunity for NIBs but does highlight the ethical issue of poorly transitioned threats to employment.

54 Medical diagnostics has been chosen to represent AI category 1A, rather than the more obvious candidate of the use of AI to profile individuals to tailor products to satisfy their needs, as this has already been discussed at length in the EIAG's *Big Tech* report.[39]

55 We will describe the operative principles and social impact of each application before going on to consider how the use of AI in the application bears upon the EIAG's framework of principles.

39. EIAG, *Big Tech*, ¶57–92.

Finally, we will consider how any tensions between the positive and negative impact on the values might be resolved, if at all.

FOCUS AREA #1A (ANALYTICAL AI, IN NON-AUTONOMOUS DECISION-MAKING MODE)
Selected example: **Medical diagnostics**

Operative principles and social impact

56 The ability to diagnose pathology (e.g. disease or trauma) is key to successful and cost-effective healthcare: the earlier and more accurately a diagnosis can be established, the greater are the chances of successful treatment and/or prevention of the onset of illness. While computers and algorithms have played a large part in medicine for many decades, the role of AI in particular has steadily grown, and in some cases has been revolutionary.

57 Three application areas are of particular note:

- The growing use of smart devices (e.g. phones and watches) to monitor human health in which the user may be alerted to medical irregularities (e.g. an abnormal heart rhythm) deserving further attention. The algorithms involved are likely to have been trained using aggregated user data and offer only "information" rather than definitive advice, thereby avoiding onerous regulatory classification as medical devices. The user is usually warned that they should seek professional medical advice before acting in any way upon the alerts presented.

- The use of AI-powered Clinical Decision Support Systems to assist healthcare professionals in making correct diagnoses, particularly in difficult cases such as rare diseases or unusual complications. Again, the software is not designed to take the place of the professional, but to offer only suggestions that then require further follow up.

- Perhaps the most widely adopted (and revolutionary) example of AI in medical diagnostics is its use in image analysis. Here, AI machine learning algorithms have been previously trained with large numbers of correctly identified abnormal features (e.g. cancerous cells) hidden within CT, MRI or microscope images. Then, when presented with new cases, these algorithms are likely to make accurate and reliable identifications, saving much time on the part of the laboratory specialist, radiologist or clinician. The resulting annotated images showing cellular or tissue segmentation into "normal" and "diseased" areas do not themselves trigger any automatic course of treatment but are used by the patient's doctor, along with other sources of information, to make a recommendation to the patient for further diagnosis and/or treatment.

Relation to Christian values

58 It is clear that AI has made a very significant contribution to human healthcare in the last few decades, speeding up research in the development of vaccines, enabling personalised medicine, and here in the case of diagnostics, increasing the accuracy and speed of diagnosis to enable more timely intervention. As such, AI-powered diagnostics can serve the common good and human flourishing by improving the outcomes of many treatments by virtue of their earlier implementation, and by freeing up clinician time to either treat more patients or spend more time with impacted patients. The revolution in AI-enhanced medical diagnostics has become an affordable best practice available to clinicians across the world, disproportionately helping patients in least-developed and developing countries. For individuals, anxieties associated with pending diagnostic investigations may be reduced by facilitating faster diagnosis, with the additional benefit of greater analysis.

59 To the extent that AI-powered medical diagnostics simply augment rather than replace the roles of the healthcare professional and patient in making judgments and decisions about courses of treatment, there would seem to be little cause for ethical concern. After all, such applications of AI could be seen to be merely extending the effectiveness of technologies that are already well established. We do not object to a surgeon relying heavily on technologies such

as X-Ray or MRI images to decide whether we need an operation, so why should we object when those technologies are made more accurate or efficient?

60 It is, however, possible to over-rely on technology, or to over-rely on one particular technology at the expense of others, or at the expense of other sources of information relevant to decision making. In this respect, AI is no different from any other technology. That being said, the way in which AI-powered medical diagnostics are presented and implemented could encourage a careless reliance in which the data are never questioned or never supplemented with other data sources. Hypothetical examples of this might be graphical representations of cancer that lack any indication of uncertainty or imply that a diagnostic decision has already been made on behalf of the healthcare professional.

61 Since universal access to good healthcare is paramount to a flourishing society and to addressing societal inequality, any breakthroughs in AI-powered diagnostics that would tend to favour some groups over others should raise serious questions and, given the challenges of data equity, those adversely affected are likely to be already marginalised groups. An example might be the inappropriate training of AI algorithms for skin cancer diagnosis using data from white-skinned people only, making diagnosis of people with darker skin less reliable and potentially with greater adverse outcomes.[40]

62 As with all AI, the automation of what was once the work of a highly trained and knowledgeable human expert could lead to the atrophy of the expertise that originally went in to training the system, as humans no longer need to gain an expertise that is available in a machine. As well as diminishing the role of experts, it risks undetected flaws being introduced by later generations of the system. The principle of humans working with AI, each having things to bring which the other cannot by itself, could be considered an important safeguard to ensure AI serves human flourishing and the common good in the long term.

Balancing the tensions

63 It is important to stress that these concerns are hypothetical: we have no evidence that current medical diagnostics are misleading users or being trained inappropriately. In practice, when it comes to medical products that make diagnostic claims, manufacturers must satisfy stringent safety and regulatory standards before the products can be launched. These standards address not just sensitivity (correctly identifying those with the disease) and specificity (correctly identifying those without the disease), but also usability (correctly using and interpreting the product).

64 If anything, greater ethical unease may lie with the use of AI in unregulated consumer products for the so-called “worried well”. It is generally regarded as a good thing for people to take an active interest in their health; however, if the tools needed to do this are available only to the wealthy, then societal health inequality might persist.

65 If we can assume that international medical device regulations are adequate to the task of ensuring safety, efficacy and usability, then AI-powered medical diagnostics may well raise very few ethical questions.

66 Broader questions, however, do remain, such as whether regulatory bodies are fully cognitive of the unique risks posed by AI, and whether government reimbursement and health economic bodies are attuned to questions of social justice.

67 Practical considerations for companies in this area:

- Are the current medical diagnostic regulations into which your product falls adequate to address any unique safety, efficacy or usability risks arising from the incorporation of AI?

40. James Zou and Londa Schiebinger, “Design AI so that it’s fair”, *Nature* 559 (2018): 325.

- Is your product or technology disruptive to the market in such a way that it might exacerbate societal inequality?

- If the AI algorithms in your product have been trained with patient data, have those patients given their explicit informed consent to their data being used in this way?

- Does the system explain its reasoning and the factors that led to the evaluation?

- Are the outputs of the diagnostic presented in such a way as to avoid the appearance of (a) 100% certainty, and (b) a decision or recommendation? To put it another way, are there sufficient warnings about the reliability and sufficiency of the output?

FOCUS AREA #1B (ANALYTICAL AI, IN AUTONOMOUS DECISION-MAKING MODE) Selected example: **Mortgage and insurance financial products**

Operative principles and social impact

68 Some kind of customer prioritising or filtering has always been an acceptable business practice up to a point: customers who represent too high a risk can legitimately be denied a mortgage or insurance product as long as the grounds for rejection do not directly or indirectly contravene anti-discrimination laws (e.g. in Great Britain the nine "protected characteristics" defined by the Quality and Human Rights Commission).[41]

69 The process of prioritising and filtering large volumes of applications incurs a significant cost for providers of these financial products, so it is understandable that they should want to use technologies capable of automating the process, and that other companies should seek to develop and market such technologies.

70 Historically, customers have supplied certain information about themselves (e.g. health status, occupation and salary) in the full knowledge that the information has a bearing on their ability to pay premiums or their likelihood of needing to make a claim, accepting that it is therefore a reasonable thing to divulge this information to a potential business provider.

71 Such information can be correlated with the profit made from each customer which can then be used to assess the probability that a new applicant will be sufficiently profitable to warrant admission. All this can be (and has already been) achieved with computers, databases and statistics programs.

72 AI is now being used to replace simple statistical programs to make the customer profiling even more accurate and reliable. However, the reasons for rejection need to be auditable. If AI is continuously learning against data, it may evolve to discriminate against some areas or sectors that perform worse, resulting in redlining or similar discriminatory practices, which are illegal.

73 The increased accuracy arises from two factors:

- The use of big data sets that go beyond what the customer has voluntarily provided (and may think is relevant) to include other socioeconomic, political, cultural and religious information harvested from various sources such as social media postings, public notices—much of which the customer may think is irrelevant and inappropriate to use.

- The ongoing learning of the AI algorithms, which adapt to the changing data set and constantly refine their predictions for any individual potential customer.

74 The increased accuracy and reliability of the customer profiling may lead product providers to dispense with a human decision-maker altogether and to allow the AI program to make the decision and inform the customer as to whether their application for a mortgage or insurance product has been successful. This is the scenario assumed here.

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41. For a good introduction, see Michelle Seng Ah. Lee, Luciano Floridi and Alexander Denev, "Innovating with confidence: embedding AI governance and fairness in a financial services risk management framework", in *Ethics, governance, and policies in artificial intelligence*, ed. Luciano Floridi (Cham, Switzerland: Springer, 2021).

Relation to Christian values

75 A higher degree of accuracy as to likelihood of default on a mortgage may help people to avoid taking on more debt than they can afford, a concern highlighted in the EIAG's *High Interest Rate and Unregulated Lending* advice.[42] If the rationale for higher premiums or rejected application is made available, this could encourage individuals to take steps to improve their financial resilience, to the extent that they are able and with signposting of support.

76 If AI is used to tailor insurance to each consumer's risk profile, then rather than rejecting an applicant *tout court*, it may be possible to offer bespoke coverage for some risks at an affordable premium, facilitating greater flexibility for marginalised groups.

77 Better accuracy means greater profitability for the business, contributing to economic growth. However, it is worth noting that too much personalisation might be less profitable than risk sharing since, by personalising, one is reducing market volume. In response, companies might need to increase price (i.e. interest rates/ premiums) to recoup for lost volume, which might then be a deterrent to personalisation.

78 However, when AI makes a negative assessment of a potential customer leading to a refusal to offer a mortgage or insurance product, the fact that: (a) a human being has not communicated the refusal; (b) there has been no opportunity for dialogue or appeal; and (c) that the reasoning behind the decision is not transparent, could all contribute to a sense that the applicant is being treated as an "it" rather than a person. This, over time, could undermine individual flourishing when people are viewed purely as economic agent.

79 There is a possibility that datasets used to train financial AI algorithms contain biases that lead to incorrect and unfair discrimination between potential applicants who would otherwise be indistinguishable, based upon their financial and medical declarations. If the final decision were left to a human, other biases might still exist, but at least there would be the opportunity for dialogue, persuasion and expression of mutual respect.

80 The tailoring of financial products to the personal risk profile of clients could lead to societal inequality by making them affordable only to the richer (and healthier) minority, and reinforcing the marginalisation of particular groups.

Balancing the tensions

81 It is theoretically possible to achieve greater human flourishing (in person or in relationship) and greater serving of the common good by harnessing the gains to be had from AI without incurring the costs. At the very least, in marginal cases where, in order to make a decision, an AI would need to "dig deeper" into the personal data of applicants, it would be preferable if this last step were performed by a human finance professional.

82 Practical considerations for companies in this area:

- What steps have been taken to ensure that data used to train the AI algorithms is representative of all potential customers? And what steps have been taken to ensure that owners of such data have given their informed consent for their data to be used in this way?
- Are your customers aware of how AI will be used in the decision-making process affecting their applications?
- Are customers given an explanation of why their applications were rejected? (this might require the AI algorithm to be capable of self-auditing).
- Is there a clear appeal or redress policy or process when customers are dissatisfied with a decision taken by AI?

42. EIAG, *High Interest Rate and Unregulated Lending: The Advice of the Church of England Ethical Investment Advisory Group,* available at: http://www.churchofengland.org/eiag/policies.

- From the perspective of Corporate Social Responsibility, are the marginal gains to be had from AI's ability to make fine-grained assessments of customer suitability justifiable in the light of any loss of societal cohesion that might result? That is, rather than segment price to the n^{th} degree, is it better to pool risk and prices beyond a certain level?

- What guardrails are in place within prompts to ensure that AI does not go beyond its remit?

FOCUS AREA #2A (GENERATIVE AI, IN STATIC/ ONE-WAY MODE)
Selected example: **AI-generated art and scripts**

Operative principles and social impact

83 Generative AI models are trained with large textual, graphical or acoustic datasets from which they discern patterns that can be used to create new data having similar characteristics. They are also trained to recognise and respond to linguistic instructions, generating apparently original works of composition on command, whether these be essays, poems, paintings, music or photographs.

84 Although the results can seem impressive to the untrained eye or ear, some would claim that the results are a pastiche that is entirely parasitic on previous human creativity and achievement, exploring the edges of what has been done before, or recombining it in new ways, rather than making any really novel transformations. Others would say that the same applies to most new art created by human beings.

85 Be that as it may, AI-generated art and scripts have been known to fool experts[43] and can certainly be good enough to pass school and university exams should candidates be tempted to cheat. Furthermore, it is only likely to get better as the algorithms improve, though this might be dependent on experts being able and willing to articulate what makes a truly genius work of art as opposed to a mere great imitation.

86 There are also serious concerns about intellectual property ownership and the use of copyrighted material to train AI data.[44]

Relation to Christian values

87 As a rapid means of pulling together all available information on a topic and presenting it in well-structured eloquent prose, generative AI can offer societal benefits in terms of cost and efficiency. It takes time and effort to collate information for a report and to précis it in a style and format suitable for its intended readership. Recent research suggests that ChatGPT can boost productivity by reducing the time to complete tasks and improving the quality of the output.[45]

88 It is possible, too, that the pedagogical potential of generative AI (e.g. demonstrating poetry in the style of X, or music in the style of Y) might be exploited to enhance learning experience and educational success, particularly for those marginalised within traditional education systems. People learn in different ways, but for many, being able to dial up a specific genre or style might aid understanding more rapidly than being given just a few examples. Generative AI language translation can also help immigrants to engage better with host societies.[46]

89 A problem with generative AI in static mode (compared to dynamic mode, below) is that its data outputs can be hosted on the same physical

43. German artist Boris Eldagsen claimed that judges of a prestigious photo award failed to distinguish between a photograph and an AI-generated image in 2023. BBC News, "Sony World Photography Award 2023: Winner refuses award after revealing AI creation", available at: https://www.bbc.co.uk/news/entertainment-arts-65296763.

44. See legal action taken by *The New York Times* against OpenAI and Microsoft for unlawful use of *The Times'* work to create AI products, available at: https://nytco-assets.nytimes.com/2023/12/NYT_Complaint_Dec2023.pdf

45. Shakked Noy and Whitney Zhang, "Experimental evidence on the productivity effects of generative artificial intelligence", *Science* 381 (2023).

46. Taking inspiration from the babel fish in *The Hitchhiker's Guide to the Galaxy*.

media used by human artists and authors, such as words in books, paint on canvass, or music on CD. There is no way of interrogating the data in real time to check its provenance, hence there is enormous scope for deliberate deception. For example, the wrongful use of AI in answering coursework or exam questions. While the temptation to cheat and defraud is not new, generative AI will make it a much more attractive proposition for some. The consequences of this are twofold, since there will be costs both to the deceived and the deceiver. For example, the deception may undermine the value of degrees or certifications and result in a loss of ability to research, reason and write. Alternatively, where someone is hired on the basis of a fake piece of writing, an employer may soon discover that the person is not up to the job.

90 Whilst enabling greater productivity, generative AI will create job disruption in those industries where this is likely to be most applicable, and will require retraining for many. There are concerns that generative AI might make some jobs obsolete, creating groups that are economically marginalised by AI.

91 Related to the above is the potential for AI to be used as a tool for misinformation (unintentional pollution of information space) or disinformation (malicious and intentionally misleading information), even at a geopolitical level. With the global surveillance and profiling of people afforded by their social media usage, individuals, political organisations and nation states can target people with information designed to sway their opinions and influence their voting.[47] Less subtly, this can include use of AI to generate so-called "deep fake" videos or photos designed to discredit or humiliate political opponents.[48] Such overt control over people's source of information goes well beyond what might be called "nudge" and is not conducive to serving the common good. In fact, some fear it could one day lead to an "infocalypse", the point when society can no longer manage a torrent of unreliable and misleading information that is difficult to distinguish from the faithful and genuine.[49]

92 More fundamentally, unlike technology designed to relieve us of the drudgery and painful aspects of some kinds of work, generative AI can relieve us of the opportunity to think or be creative. Quite apart from the risks of deception, herein lies an even greater long-term danger: that by lazily relying on the click of a mouse to produce computer-generated art, we begin to forget how to think constructively, imaginatively, and non-algorithmically. If we allow these skills to atrophy, we effectively disinherit an important God-given means for individual flourishing: the ability to create.[50]

93 As generative AI adds its own results to the internet, and as these results become part of the training of future generative AI programs, some commentators have expressed concern that increasing reliance on generative AI to source knowledge will lead to a degrading of the internet as it becomes populated by content that is derivative rather than original.

Balancing the tensions

94 One could argue that since this application of AI has not arisen in response to any pressing societal need but has the potential to encourage the vices of bearing false witness and sloth, we should treat it with some ethical suspicion at best. Saying that, many technologies have begun life as solutions to new needs invented by their makers but have then gone on to offer unforeseen benefits. A classic example is the Sony Walkman, originally created so that Sony's co-founder could listen to music on long flights, but which revealed to the world the possibility of affordable portability and privacy in an electronic device.

47. See for example the alleged Russian meddling in the 2020 US election.
48. A recent example being a clip of Nancy Pelosi edited to make her look ill and impaired.

49. Mustafa Suleyman, *The coming wave: AI, power and the twenty-first century's greatest dilemma* (London: The Bodley Head, 2023), p. 172.
50. This might be expressed theologically in terms of being made in the image of God (the *imago dei),* according to which (in some interpretations) we are God's created co-creators. See, for example: Philip Hefner, "The evolution of the created co-creator", in *Cosmos as creation: theology and science in consonance*, ed. Ted Peters (Nashville, TN: Abingdon Press, 1989).

95 While there is no denying that, in certain circumstances, generative AI might be a useful short-cut to producing something of acceptable value under tight time and cost constraints, this value is easily over-stated and can instead be interpreted as a lack of contentment with our usual embodied means of skill acquisition and practice. Rather than being driven by a clear disvalue (e.g. suffering due to sickness, old age or poverty) it is driven by the naïve attitude that any effort or self-discipline is something to be avoided and overcome.

96 On the other hand, it tends to be the case that technology does not obviate the need for human effort and skill but simply takes it to a different level. Just as there is a good way and a bad way to use an axe or a chain saw, so too there is a good way and a bad way to use a computer, the internet or an AI program. The fact that AI has democratised certain ways of achieving a result does not remove the need for skilled and knowledgeable people who know how to use AI in a way that will produce a good quality result (e.g. in avoiding or detecting "hallucinations" introduced by generative AI[51]).

97 Practical considerations for companies in this area:

- Do the data sets with which the generative AI algorithms have been trained contain any confidential or proprietary information? If so, have the necessary legal permissions been received?
- What steps have you taken to avoid the AI being used for nefarious purposes? e.g. does the algorithm decline to execute requests that might be unlawful, inciteful or divisive, such as supplying answers that are racist or homophobic, or expressing views that are politically biased?
- Are there means to audit the output of your algorithms to detect that AI has been employed (e.g. watermarks or metadata), and to reverse engineer the sources of the information?
- To avoid lulling people into a false sense of security, does the generative AI express self-doubt, solicit feedback, and provide citations or interrogable evidence in support of its conclusions?

FOCUS AREA #2B (GENERATIVE AI, IN DYNAMIC/TWO-WAY MODE). Selected example: **Robotic healthcare companions**

Operative principles and social impact

98 A global ageing population and a shortage of healthcare professionals has created a new opportunity for AI in the form of conversational assistants. This is particularly the case in Japan, where an increasing number of people are living alone and a significant number are dying so-called "lonely deaths" *(kodokushi),* with their bodies discovered a long time after their deaths.[52]

99 These may be virtual assistants having voice only (a chatbot), or embodied assistants in anthropoid or nonhuman form, equipped with senses, facial expressions, postures and other bodily movements, as well as voice.

100 The part played by AI algorithms can be anything from providing information in response to verbal requests, to holding sustained ad lib conversations, all the way to detecting the mood of the patient and offering emotional support.

101 Accurate simulation of human appearance is not required and can even be counter-productive due to a phenomenon known as the "Uncanny Valley" (Masahiro Mori) in which human affinity for a robot undergoes a sharp downturn as perfect human likeness is approached but not achieved.[53]

51. For more on "hallucinations" see: IBM, "What are AI hallucinations?", available at: https://www.ibm.com/topics/ai-hallucinations.

52. Nils Dahl, "Governing through kodokushi. Japan's lonely deaths and their impact on community self-government", *Contemporary Japan* 32, no. 1 (2020).
53. Masahiro Mori, "The uncanny valley", *IEEE Robotics & Automation Magazine*

102 What is missing in the simulation may be compensated for by a degree of anthropomorphising or psychological projection on the part of the patient, in the same way people might talk to their pets as if they were human persons, even though they know they are not.

103 Examples:

- The Woebot mental health tool, hosted on a smartphone.[54]

- Paro, the interactive robot in the form of a white furry seal, designed for dementia patients. Clinical studies have shown it is effective in improving mood, reducing anxiety, boosting sleep and reducing pain perception.[55]

104 The probabilistic and partially autonomous nature of neural networks makes them ideal algorithms for simulating human relationships. We do not expect other human beings to be correct, accurate, or logical all the time, and therefore some degree of quirkiness or unpredictability may help give the patient the feeling that they are talking to a real human person. However, there is a big difference between human idiosyncrasies and the totally out of the park responses current AI algorithms can give.

105 In theory, the robot could be pre-programmed with a knowledge of the patient's background and preferences such that they can more quickly develop a relationship with the patient, though greater authenticity might be better achieved with a slow "getting to know you" process.

106 Chatbots and robot companions can also use AI to diagnose medical conditions and relay them to healthcare professionals, especially if they are equipped with visual, acoustic and tactile sensors. Smart watches do this to some extent already.

107 Ever since the ELIZA program of 1965, it has been known that patients will open up more to computer programs than to real human beings (which raises data protection issues if the conversation is recorded and/or made available for other uses).

Relation to Christian values

108 The ignominious decline and death of many neglected elderly people is the very antithesis of flourishing as persons. Anything that AI can do to help mitigate this evil must surely be welcomed. One could argue that it is better that society send an AI surrogate than to send no-one or nothing at all.

109 More specifically, robotic healthcare companions can support individual flourishing in various ways, such as: being able to summon help quickly if physical or mental health declines; providing reminders to take medicine; and augmenting failing memory both proactively ("remember Joe is coming today"), and reactively ("what was that programme I was watching earlier?").

110 Those with profound communication difficulties can already be helped by technologies such as eye-following and speech generation. Improved AI capabilities to "read" a patient's intent (from brain implants or facial expressions), and to project a patient-specific voice or avatar in conversation with others, could significantly improve their relationships with others.

111 While robots clearly perform worse than humans in terms of the empathy and benevolence needed to allow patients to flourish in relationship, they can and do surpass most humans with their endless patience, constantly availability and non-judgemental listening,[56] and

(June 2012), available at: https://doi.org/10.1109/MRA.2012.2192811.

54. See Woebot Health, https://woebothealth.com/.

55. See Paro Therapeutic Robot, https://www.paroseal.co.uk/.

56. Diana Löffler, Jörn Hurtienne, and Ilona Nord, "Blessing Robot BlessU2: A discursive design study to understand the implications of social robots in religious contexts", *International Journal of Social Robotics* 13 (2021).

can support marginalised groups in situations where real humans may fail to stand with them.

112 Robots can be programmed and trained to detect the outward display of emotions of a human being and respond in an empathetic way, but they are incapable of feeling in response to another human. This simulated empathy would be considered sociopathic in humans, presenting the danger that "empathetic" healthcare robots will seem manipulative and empty, thereby limiting the extent to which a patient can flourish in relationship. Herzfeld points to another limiting factor in such relationships: "Aging is a grace not given to robots. How would it feel for a person to age while their robot does not?"[57]

113 We do not properly stand with the marginalised or serve the common good when we adopt an insincere or arm's length approach to healthcare and convalescence. Nor do we ourselves flourish as persons or in relationship when we allow robots to deputise for our presence. Authentic solidarity is important to the emotional and spiritual formation of the healthy as well as the sick. It is one thing to augment our human abilities with technology, but to replace ourselves entirely in our relationships with others is to risk downgrading our humanity. As less is given to another, less will be expected from the other, until we forget what we had to give and receive in the first place.

114 As Herzfeld has asked: "Mimicry is a two-way street…If we surround ourselves with social robots, would we copy them, perhaps reducing our own expression of emotion, in the same way that the widespread use of Twitter has, for many, reduced their style of verbal communication?"[58]

115 There are concerns that, with dementia sufferers in particular, robots will be mistaken for real human beings and their limitations in that capacity will only seek to accelerate cognitive decline rather than enhance it (through infantilisation).[59]

Balancing the tensions

116 As with most technologies, it is quite possible with AI to achieve a happy medium between complete disallowance and total dependence. Though some might argue that only a complete rejection of companion robots can avoid a slippery slope from augmentation of human company to permanent replacement, others might claim that a certain level of reasonable use of such technology is actually better than constant use or no use at all. When frail elderly people are living alone or in care homes it is impossible for friends and relatives to be permanently present, and when they are absent, the compensating presence of a robot—just as with an animal pet—might be psychologically more beneficial than having access only to TV, radio and social media (in fact this raises an interesting question as to the additional benefits accrued in making patients less reliable on social media, which carries its own dangers).[60]

117 How it might be possible to stop such technology from becoming a permanent alternative to human visitors is difficult to say. One option might be to stop short of making the robot too good in its companionship skills, to have built in features that remind the human owner that the robot is not a human being, or to have "animaloid" rather than humanoid form.[61]

118 There is also need for further research to assess the psychological effects of sustained exposure to realistic companions before any attempts are made to make them even more realistic.

57. Herzfeld, *The artifice of intelligence*, p. 170.
58. Herzfeld, *The artifice of intelligence*, p. 118.
59. John Wyatt, "The impact of AI and robotics on health and social care", in *The robot will see you now: artificial intelligence and the Christian faith*, ed. John Wyatt and Stephen N. Williams (London: SPCK, 2021), p. 188.
60. See the dangers narrated in: Sherry Turkle, *Alone together: why we expect more from technology and less from each other* (New York: Basic Books, 2011).
61. For more on the dangers of anthropomorphising robots, see: Darling, "'Who's Johnny?' Anthropomorphic framing in human-robot interaction, integration, and policy".

119 Practical considerations for companies in this area:

- Is the robot too realistic and in danger of deceiving patients who are cognitively impaired?
- Are there safeguards in place when the users/patients are very vulnerable and have diminished cognitive capacities?
- Does the robot record private dialogue and share it with third parties, and if so, what restrictions or opt-outs are in place?
- Is the robot designed to bring the best out of people, stimulating and stretching them, or is it programmed to appeal to their baser instincts?
- What safety features are built into the robot's design to avoid physical or emotional harm being done to the patient?

RECOMMENDATIONS

120 The five theological principles used throughout this report are intended to guide our thinking about companies using and developing AI technologies, but also the behaviour of other organisations, including the NIBs themselves, and governments. While the principles are theologically grounded, they are ethical principles or values that are relevant to everyone: their force is moral, and they retain their moral force even if they may also need to be implemented in legislation, regulations or codes of practice to be more fully effective.

121 There is a clear case for investment in AI and the benefits it can bring to society and human flourishing through rapid advancements in healthcare and climate solutions, amongst many others, provided this is done responsibly and that the benefits of AI are distributed fairly. This advice on AI was borne out of the EIAG's *Big Tech* report, and we reaffirm that report's recommendation that companies take responsibility for their use of AI and its development.[62] The EIAG expects ethical considerations to be integrated into the full lifecycle of AI, from research and development, testing, through to its dissemination and use by companies when purchasing off-the-shelf or bespoke AI applications, to ensure that AI can make a positive contribution to the long-term good of society as a whole. Although this report has focused on existing and feasible near-term AI technologies, it is expected that ethical considerations be integrated also into the design and development of ambitious long-term AI technologies.

122 We also reaffirm the four commitments expected of Big Tech companies in this report, and believe they should be extended to companies developing and using AI:[63]

a. a commitment to verifiable transparency,
b. a commitment to promote human-centred design,[64]
c. a commitment to enable the flourishing of children and other vulnerable groups,
d. a commitment to foster a tech ecosystem that serves the common good.

123 The EIAG supports the principles of the Rome Call for AI Ethics, endorsed by General Synod in 2024, and these are aligned with the Group's thinking. However, the purpose of this paper is to advise the NIBs on matters of ethical investment in relation to AI, and therefore simply to recommend the Rome Call would not be sufficient to fulfil this purpose and more in-depth and specific advice which considers this through the lens of investment is needed to support the NIBs.

124 There are a number of challenges to be faced in framing practical ethical investment recommendations for the NIBs in this advice. The sheer breadth and pervasiveness of AI

62. EIAG, *Big Tech*, ¶128–39.
63. See EIAG, *Big Tech*, ¶138.
64. By this is meant a design that takes into account safety, privacy, freedom and fairness, such as allowing the user to opt out of certain functions, providing informed consent if their data is being used, etc. That is to say, beyond just ergonomic design intended to avoid human error.

applications and their use within companies makes it difficult to isolate any potential ethical issue from wider business operations. In addition, the speed at which AI is developing, and the difficulty in foreseeing what technologies or applications might emerge, necessitates the principles-based approach found in these recommendations.

125 Given the breadth of the applications and impact of AI already highlighted in this paper, it is important to note other relevant EIAG advice that the NIBs can draw on to underpin their approach to these challenges. In particular, EIAG advice on human rights, environment and defence can support the NIBs in addressing intersecting issues such as the environmental impact of AI training, use of AI in weaponry, and the broad spectrum of human rights concerns relating to AI.[65]

126 The following recommendations indicate some of the lines that investors may wish to pursue. We recognise that it may not be fully clear yet what AI fully oriented to prioritising human values would look like, which is why it is important to keep the EIAG's five theological principles in mind. We recommend to the NIBs a proportional approach to addressing questions about the use of AI in their holdings, prioritising those questions for industries and applications that present the greatest risk and impact to individuals. We also recognise that priorities for engagement will change as opportunities occur, as global norms on AI become more established and as companies change their policies and practices over time.

A Corporate Culture of Responsible AI

127 The corporate culture of a business is the means by which it expresses virtue, and can be the starting point for responsible action in relation to AI. For companies using and deploying AI within their business, we expect the adoption of these systems to be explored and approached in a culture of responsibility and transparency. AI should be recognised as a tool that can serve business, but it is essential that companies recognise the potential scale of negative impact that AI systems can have when not developed and deployed responsibly. As far as possible, use of AI systems should enable workers to learn more and to work more efficiently and effectively.

128 The integration of AI systems into business activities should be strategic and well-governed, with executive oversight for AI risks and ethics. Clear guardrails should be in place to mitigate risks, and should include means of audit, appeal and redress, with a plan for how these are operationalised at all levels. There must be a means of auditing the outcomes of AI systems to ensure that these are accurate and do not discriminate on the grounds of "protected characteristics". Appropriate training should be given to all those with responsibilities for using or overseeing use of AI systems, so that workers are sufficiently trained to use this technology correctly and effectively, and that those with oversight responsibilities have the right knowledge to assess and mitigate risks when procuring and deploying AI systems.

129 AI systems deployed in a business context should respect and protect human rights, and their training and deployment should be fair and inclusive. The adoption of AI systems should be undertaken in consultation with relevant stakeholders at all levels. Where the adoption of AI systems is likely to result in job replacement or displacement, this should be considered in consultation with all stakeholders, and with a clear, long-term strategy for adoption, implementation and transition of systems and workforce, alongside meaningful opportunities for reskilling.

Establishing Standards of Practice

130 As this paper has already noted, common ethical concerns and corresponding principles of action are beginning to emerge in a plethora of secular frameworks on AI ethics. In order to support both developers of AI and companies deploying and using AI in their operations

65. See EIAG, *Human rights;* EIAG, *Climate Change;* and EIAG, *Defence Advice,* all available at: http://www.churchofengland.org/eiag/policies.

to take responsibility for the impacts and risks associated with these systems, there need to be coordinated efforts to establish industry standards of practice for the whole AI lifecycle and, where necessary, development of appropriate policy and regulation. The EIAG recognises that national and international policies are a key means by which to encourage and manage responsible practices and an area in which investors have influence.

131 Echoing the four commitments in the *Big Tech* report, the EIAG recommends that the NIBs engage to encourage policy, codes of practice and corporate commitments in line with the following principles:

Human-centred design and deployment

132 Central to the EIAG's theological and ethical approach is the impact of AI on human beings and the importance of the protection of that which is distinctly human. As Pope Francis described in an address on AI, "At this time in history, which risks becoming rich in technology and poor in humanity, our reflections must begin with the human heart."[66] AI technology is fundamentally a tool for human beings, one that can be used for benefit or for harm. Like all good tools, AI systems should enhance human flourishing, both in the purposes for which they are used and in the expertise or craftsmanship that is required to use them well.

133 Companies involved in any part of the lifecycle of AI have a responsibility to ensure that they protect human rights and the AI does not violate these at any stage.[67] AI should not perpetuate bias and discriminate against people based on "protected characteristics", and training data should be geographically and socially representative. The use of synthetic data in training AI models should be considered carefully and, where this is necessary, higher levels of due diligence and regular auditing should be carried out to ensure that synthetic data outputs are reliable and do not perpetuate bias or misrepresentation.

134 Where AI is deployed in a company, managers should ensure employees are given appropriate training to use AI and, where this may make jobs redundant, offer fair and equal opportunities for high-value upskilling and reskilling. Workforce planning for the displacement of jobs due to AI adoption needs to take place at government level as well to identify the sectors likely to be most affected. Such planning should include initiatives for upskilling and reskilling current workers and education and training where it is needed for a future workforce.

135 Where AI is used appropriately in decision-making, this must include human verification and determination, i.e., "a human in the loop", with means of auditing AI outputs and of appeal and redress for incorrect or biased outputs.

Accountability

136 Company leaders developing and deploying AI should take responsibility for ensuring the safety and security of the AI technologies that they either develop or outsource, and be held accountable for any malfunctioning algorithms. The EIAG expects these companies to consider the full impact of AI development, deployment and use—including environmental concerns such as energy usage—and to demonstrate a commitment to rigorous standards of AI governance that go beyond merely meeting minimum standards throughout the whole system lifecycle.

137 Companies developing AI should clearly define the purpose of the AI system in development and integrate senior management level accountability for responsible development, training and deployment of AI. Developers should also ensure guardrails are in place to prevent AI systems going beyond their intended remit, and have mitigations to prevent AI being used for unethical or nefarious purposes.

66. Vatican, "Message of His Holiness Pope Francis for the 58th World of Social Communications" (24 January 2024), available at: https://www.vatican.va/content/francesco/en/messages/communications/documents/20240124-messaggio-comunicazioni-sociali.html
67. Further reflection on the duty of businesses to respect human rights can be found in the EIAG's *Human rights* advice.

138 Companies deploying AI should clearly define why and how they are using AI and consider whether AI is necessary to fulfil the task. They should take responsibility for the outcomes and use of bespoke and off-the-shelf AI systems that they deploy, and ensure that procurement, due diligence and audit processes are sufficient to assess and manage AI risks. At a senior management level there should be accountability for AI strategy and implementation, including ethics and risk management, and there should be a framework for how this is operationalised at all levels.

Transparency

139 Companies developing AI should be transparent about the sources of data used for developing and training AI, which should not include data which is subject to copyright (without obtaining appropriate license), or personal or confidential data where explicit consent has not been given.

140 AI should be explainable and auditable, including the rationale for decision making. Companies deploying AI should be transparent about when and how they are using AI, including signposting means of appeal and redress when AI is used in decision making.

141 Products of generative AI systems should contain means to verify whether AI has been employed in their creation.

Characteristics of Unethical Uses of AI

142 Given the scale at which AI is developing and its potential applications and uses, it is not necessarily possible or helpful to provide the NIBs with a list of applications of AI that the EIAG considers to be unethical. Rather, we have identified characteristics of uses of AI by companies that may be considered unethical, on the basis of the five theological principles and grounded in the EIAG's *Human rights* advice.[68] This is intended not as an exhaustive list, but to aid the NIBs in identifying those companies or industries within their holdings that may warrant further due diligence and a means of engaging on this basis.

- High stakes autonomous decision making that have irreversible impact on an individual's safety, rights or wellbeing (including life or death decisions, in the case of AI in weaponry)
- Uses in which AI replaces, and therefore devalues, human relationships without explicit consent and awareness, particularly with respect to children and other vulnerable people
- Unjustified and disproportionate surveillance, including mass surveillance and social scoring
- Use of generative AI to cause harm to others, including discrimination based on "protected characteristics"
- Use of AI to create and spread misinformation and disinformation
- Manipulating an individual's behaviour, without explicit knowledge or consent for the use of AI in a particular service or application
- Use of AI where the environmental impacts of training AI models is not proportionate to the value of the output

143 The EIAG encourages the NIBs to give particular attention and appropriate monitoring or due diligence to those industries where there is a higher likelihood of exposure to uses of AI with these characteristics, including financial services and insurance.

144 NIBs should be aware of the *potential* for any new AI technology to be used for nefarious purposes, even if the developers have no intention of marketing towards such applications. Developers should be aware of the different ways their algorithms could be misused and take steps (through IP protection, licensing restrictions or technological barriers) to ensure they do not unwittingly become "accessories to evil".

68. EIAG, *Human rights.*

145 Given the lack of established global AI norms, appropriate data points to support normal investor methodologies and to engage and monitor relevant holdings are only in their infancy or do not exist at all. The EIAG also encourages the NIBs to support efforts to establish frameworks and appropriate data points to understand the nature and scale of AI usage within companies, and to ensure that AI ethics and risks are integrated into their governance and operational structures.

146 To aid investors in identifying those sectors or AI applications that would be considered high risk or unethical, examples of these can be found in Appendix 2.

RELATED EIAG ADVICE

Big Tech policy and advice
Human Rights policy and advice
Climate Change advice
Defence advice

REVISION LOG

Revision date	Sections revised	Description
Dec 2024		Advice published

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APPENDIX 1

AI markets and example applications include the following:[69]

1 Financial services

- Personalised financial planning
- Fraud detection and anti-money laundering
- Process automation, including customer-facing operations

2 Technology, communications and entertainment

- Media archiving and search
- Customised content creation
- Personalised marketing and advertising

3 Retail

- Personalised design and production
- Anticipating customer demand
- Inventory and delivery management
- In-store identification of customers for marketing or theft prevention purposes

69. Based on Price Waterhouse Cooper's predictions of where AI will have the greatest potential. See Arnand S. Rao and Gerard Verweij, *Sizing the prize: what's the real value of AI for your business and how can you capitalise?*, PwC (2017).

4 Healthcare

- Diagnostic imaging
- Continuous monitoring for early detection of disease symptoms and progression
- Clinical decision support systems
- Companion robots
- Early indication of pandemics

5 Automotive

- Autonomous fleets for ride sharing
- Semi-autonomous features such as driver assist
- Individualised route planning to reduce congestion
- Engine monitoring and predictive, autonomous maintenance

6 Transportation and logistics

- Autonomous trucking and delivery
- Traffic control and reduced congestion
- Enhanced security

7 Energy

- Smart metering
- Efficient grid operation and storage
- Predictive infrastructure maintenance

8 Manufacturing

- Increased automation, enhancing consistency, efficiency and throughput
- Enhanced monitoring and auto-correction of manufacturing processes
- Supply chain and production optimisation
- On-demand production

APPENDIX 2

Real world examples of unethical use of AI

1 Deepfakes
e.g. The DeepNude App, which transforms images of clothed individuals into a representation of what they might look like undressed.[70]

2 Racial and gender bias
e.g. Amazon's AI hiring tool, which trained itself that male candidates were preferable.[71]

3 Surveillance and privacy invasion
e.g. Clearview AI was fined by the UK Information Commissioner's Office for unlawfully storing facial images that it had scraped from the internet, though the fine was overturned on appeal due to Clearview claiming that the images would be used only by law enforcement bodies outside the UK.[72]

4 Automated disinformation campaigns
e.g. Social media bots in elections that spread political messages that are often intentionally divisive and hateful.[73]

5 Manipulative algorithms
e.g. Facebook's news feed algorithm that has been used in a secret study involving 689,000 users in which friends' postings were moved to influence moods.[74]

70. The original DeepNude app was taken offline after a backlash, but many other alternative similar apps have since sprung up in its place. See Maya Oppenheim, "App which used algorithm to 'undress' women and create fake nudes shut down", Independent (29 June 2019), available at: https://www.independent.co.uk/tech/deepnude-app-women-naked-remove-clothes-shut-down-a8980731.html.
71. See "Amazon scrapped 'sexist AI' tool", *BBC News* (10 October 2018) , available at: https://www.bbc.co.uk/news/technology-45809919
72. See Chris Vallance, "Face search company Clearview AI overturns UK privacy fine", *BBC News* (18 October 2023), available at: https://www.bbc.co.uk/news/technology-67133157.
73. See e.g. "Investigation reveals content posted by bot-like accounts on X has been seen 150 million times ahead of the UK elections", *Global Witness* (2 July 2024), available at: https://www.globalwitness.org/en/campaigns/digital-threats/investigation-reveals-content-posted-bot-accounts-x-has-been-seen-150-million-times-ahead-uk-elections/.
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